LP3 Assignment A5

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Problem Statement:
Write a program for analysis of Quick sort by using deterministic and randomized variants.

```
#include <iostream>
#include <vector>
#include <cstdlib>
#include <ctime>
#include <chrono>
using namespace std;

long long comparisons = 0;

int partition(vector<int>& a, int low, int high, bool randomized) {
    if (randomized)                          // random pivot moved to end
        swap(a[low + rand() % (high - low + 1)], a[high]);
    int pivot = a[high], i = low - 1;
    for (int j = low; j < high; j++) {
        comparisons++;
        if (a[j] <= pivot) swap(a[++i], a[j]);
    }
    swap(a[i + 1], a[high]);
    return i + 1;
}

void quickSort(vector<int>& a, int low, int high, bool randomized) {
    if (low < high) {
        int p = partition(a, low, high, randomized);
        quickSort(a, low, p - 1, randomized);
        quickSort(a, p + 1, high, randomized);
    }
}

vector<int> analyze(vector<int> a, bool randomized) {
    comparisons = 0;
    auto start = chrono::high_resolution_clock::now();
    quickSort(a, 0, a.size() - 1, randomized);
    auto end = chrono::high_resolution_clock::now();
    cout << (randomized ? "Randomized    " : "Deterministic ")
         << "| Comparisons: " << comparisons
         << " | Time: " << chrono::duration<double, milli>(end - start).count() << " ms\n";
    return a;
}

void print(string label, const vector<int>& a) {
    cout << label;
    for (int x : a) cout << x << " ";
    cout << "\n";
```

```
}

int main() {
    srand(time(0));
    int n;
    cout << "Enter number of elements: ";
    cin >> n;

    vector<int> arr(n);
    for (int& x : arr) x = rand() % 100;

    print("Before sorting: ", arr);
    analyze(arr, false);
    vector<int> sorted = analyze(arr, true);
    print("After sorting:  ", sorted);
    return 0;
}
```

Output:

```
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ g++ LP3A5.cpp -o a5
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ ./a5
Enter number of elements: 18
Before sorting: 99 43 26 40 95 31 53 21 95 68 53 35 26 66 43 40 45 78
Deterministic | Comparisons: 55 | Time: 0.002374 ms
Randomized    | Comparisons: 62 | Time: 0.003537 ms
After sorting:  21 26 26 31 35 40 40 43 43 45 53 53 66 68 78 95 95 99
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ ./a5
Enter number of elements: 15
Before sorting: 46 28 87 85 13 55 65 47 55 82 56 88 9 18 77
Deterministic | Comparisons: 42 | Time: 0.001823 ms
Randomized    | Comparisons: 43 | Time: 0.001793 ms
After sorting:  9 13 18 28 46 47 55 55 56 65 77 82 85 87 88
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ ./a5
Enter number of elements: 16
Before sorting: 81 30 95 70 27 84 72 63 23 4 81 23 59 9 44 46
Deterministic | Comparisons: 48 | Time: 0.002144 ms
Randomized    | Comparisons: 51 | Time: 0.002124 ms
After sorting:  4 9 23 23 27 30 44 46 59 63 70 72 81 81 84 95
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
```